

SRABON DEBNATH

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U.S. Citizen & Eligible for Security Clearance

Cloud/AWS Engineer with a B.S. in Computer Science (George Mason, Dec 2025) and hands-on experience building production-grade AWS infrastructure. Architected and deployed a live, highly available web environment (debnath.click) using EC2, ALB, ASG, Route 53, ACM, and CloudWatch. Solid foundation in Linux, networking, IAM, and security. U.S. citizen eligible for security clearance and targeting DC/NoVA cloud engineering roles.

SKILLS

- **Cloud & AWS:** EC2, Application Load Balancer, Target Groups, Auto Scaling Group, Route 53, AWS Certificate Manager, CloudWatch, IAM, VPC, Security Groups, EC2 User Data scripting.
- **Linux & Infrastructure:** Linux Commands, SSH, Apache HTTP Server, Web Server Deployment, File Permissions, System Services, Log Troubleshooting.
- **Networking:** DNS, HTTP/HTTPS, Ports, Load Balancing, Health Checks, SSL/TLS, Public/Private Access Control.
- **Programming & Databases:** Java, SQL, C, Spring Boot, REST APIs, Hibernate ORM, Spring Data JPA, Oracle, H2.
- **Automation & Testing:** Java Automation, Selenium, TestNG, JUnit, Functional Testing, Regression Testing.
- **Tools & Methodologies:** Git/GitHub, Maven, Postman, IntelliJ IDEA, Eclipse, VS Code, Jira, Agile/Scrum.

PROJECTS

Highly Available Static Website on AWS

Tech Stack: AWS EC2, Apache, Application Load Balancer, Auto Scaling Group, Route 53, ACM, CloudWatch, Security Groups, User Data, Linux.

GitHub: github.com/srabondeb/aws-highly-available-static-website | **Live:** debnath.click

- **Architected** and deployed a fault-tolerant web application on AWS, provisioning multiple **EC2** instances running Apache behind an Application **Load Balancer** with Target Group health checks to eliminate single points of failure.
- **Engineered** an **Auto Scaling Group** (min 2 / max 3 instances) with a Launch Template and automated User Data bootstrap script to install dependencies, pull latest code from GitHub, and serve content, enabling **zero-touch horizontal scaling**.
- **Configured Route 53 DNS** routing to a custom domain (debnath.click) and provisioned a free **SSL/TLS certificate** via ACM, enforcing HTTPS across all traffic.
- **Hardened network security** by designing layered security group rules and restricting inbound SSH to a single trusted IP while routing all public HTTP/HTTPS traffic exclusively through the Load Balancer, following AWS least-privilege principles..
- **Implemented CloudWatch monitoring** with a CPU utilization alarm (threshold: 70%) to proactively detect performance degradation and support infrastructure observability..

Network Routing Simulator

Tech Stack: Java (Multithreading, TCP Sockets), Eclipse, Git.

- Simulated a distributed network routing system in Java, implementing both Distance Vector Routing (TCP sockets, multithreading) and **Link State Routing** (Dijkstra's Algorithm) to compare convergence behavior across nodes.
- Instrumented per-round logging to track routing table updates, verify convergence, and debug distributed network state – reinforcing practical understanding of network fundamentals underlying cloud infrastructure.

Selenium Test Automation Framework

Tech Stack: Java, Selenium WebDriver, TestNG, Maven

- **Developed an automated UI** testing framework using Selenium WebDriver, Java, and TestNG to validate end-to-end web application workflows including form submissions, navigation, and file uploads
- **Enhanced test reliability** and maintainability through WebDriverWait, ExpectedConditions, and reusable setup/teardown methods, reducing flaky test failures.

WORK EXPERIENCE

Front Desk & Administration – Wyndham Garden Manassas, VA (2018–Present):

- Audited and reconciled daily transaction logs across disparate systems, identifying and resolving discrepancies with consistent accuracy and zero unresolved errors..
- Triageed and resolved high-priority operational incidents in a fast-paced environment, prioritizing by business impact and maintaining clear stakeholder communication throughout
- Partnered cross-functionally to diagnose and remediate recurring operational bottlenecks, ensuring uninterrupted service delivery across daily operations..

EDUCATION

Bachelor of Science in Computer Science

George Mason University, Fairfax, VA 22030

Graduated: December, 2025

Relevant Coursework: Data Structures & Algorithms, Database Systems, Networking, Software Testing, AI